

Software Applications for Computer Science

Topics you can represent after the midterms

Structure of this seminar

1	Friday, September 11, 2026	Quarto/ RStudio/ APA
2	Friday, September 18, 2026	Quarto/ RStudio/ APA
3	Friday, September 25, 2026	Introduction and first hands-on example
4	Friday, October 2, 2026	Operating Systems, basic security and more about CLIs
5	Friday, October 9, 2026	AI ecosystem: LLMs, tokens, agents, harnesses and more
6	Friday, October 16, 2026	Using Git, Github and Claude for Creating your own Website
7	Friday, October 23, 2026	More Time for Creating your own Website
	— — —MID TERMS— — —	After the Mid Terms it is your turn.
8	Friday, November 6, 2026	
9	Friday, November 13, 2026	
10	Friday, November 20, 2026	
11	Friday, November 27, 2026	
12	Friday, December 4, 2026	
13	Friday, December 11, 2026	
14	Friday, December 18, 2026	

Topics for your presentation

1. Git, GitHub and GitLab
2. IDEs
3. Claude Code or OpenCode
4. Figma or Claude Design
5. (PowerBI)
6. Docker and Containerization
7. Jira for project management
8. Cloud and DevOps

Updated assessment methods and grading criteria

- Probably everybody is doing a presentation alone
- Minimum duration is 10min
- Presentation has theoretical and practical part
 - Exception: Cloud and DevOps is only theoretical
 - Practical part: You show how to do a small task on your pc
 - **It is okay if something does not run as expected in practice**
- Handout is delivered as a markdown file and with terminal commands (if you used some)

Git, GitHub and Markdown

We cover the basics by setting up our own website, you can go more in depth.

- Github
 - Git vs. GitHub: tool vs. platforms
 - Fork, clone, star, watch: finding and reusing other people's code
 - Issues: tracking bugs and ideas
 - Branches and merge/pull requests: how teams review code
 - *Git workflows: feature branch workflow vs. working directly on main*
 - *Understand how multiple people can develop software together*
 - *GitHub Actions: automatic tasks*
- Markdown (the best format for AI)
 - The basics: headings (#), bold/italic, lists, links, images, code blocks, tables
 - Where you see it: README.md, GitHub issues, documentation, your handout
 - Live: create a .md file in GitHub (the handout) and do a merge request

IDEs

- What is an IDE and why not just a text editor
- Components: editor, file tree, terminal, run configurations
- Syntax highlighting, autocompletion, refactoring (e.g. rename)
- Git integration inside the IDE
- Debugger: breakpoints, stepping, variables, call stack (frames)
- Live: set a breakpoint in PyCharm, step through a loop, watch variables change

Claude Code / OpenCode

- Chat vs. agent: an agent reads files and runs commands itself
- Sessions: starting, resuming, clearing context
- Choosing models with /models (cost vs. quality)
- Modes: plan, ask, auto (how much control you give away)
- CLAUDE.md: project instructions the agent reads every time
- CLAUDE.md and Git: commit it so the whole team shares it, and review agent changes via diff and commits
- Risks: always review output, never paste secrets
- Live: let the agent make a small change in a repo, then inspect the diff

OpenRouter and OpenCode as a free alternative

- Sign in with your Github account on OpenRouter
 - Save the API key
- Run open code auth, select OpenRouter as a service, paste API key
 - */models* and type *free*
- Free models are often not available, then this error message appear
 - (Console): Upstream request failed: Endpoint is unavailable.
 - Just try another free model
- 200 requests per day

Figma/Claude Design

- Design before code: wireframe, mockup, prototype
- Figma basics: frames, components, auto layout
- Collaboration: comments, sharing, developer handoff
- Claude Design: generating a design from a prompt
- Good prompts: describe purpose, audience, style
- Live: create a design with Claude Design

PowerBI

- What is BI: turning raw data into decisions
- Typical workflow: import data, clean it, model it, visualize it
- Data sources: Excel, CSV, databases
- Visuals and dashboards: choosing the right chart
- Filters and slicers: making dashboards interactive
- *Sharing reports*
- Live: load a CSV and build a small dashboard

Docker and Containerization

- Problem: "it works on my machine"
- Container vs. virtual machine (lightweight, shares the OS kernel)
- Image vs. container (recipe vs. running instance)
- *Docker Hub: downloading ready-made images*
- *Dockerfile: describing your own image*
- Live: run a container (e.g. a web server) and open it in the browser

Jira

For Project Planing; *Italic type is optional*

- Jira: tracking work as tickets (epic, story, task, bug)
- Board, backlog, status workflow (to do, in progress, done)
- Agile (Scrum/Kanban) vs. waterfall: short iterations vs. fixed phases
- Sprint, standup, retrospective in one slide
- *Jira + GitHub: linking tickets to branches and commits*
- Pricing: free plan for small teams (please check current limits before the seminar)
- Live: create a ticket and move it across the board

Cloud and DevOps

No practical part here

- Cloud: renting computers, storage and services instead of buying hardware
 - Service models: IaaS, PaaS, SaaS
 - Big providers: AWS, Azure, Google Cloud
- DevOps: development and operations as one team
 - CI/CD pipeline: code, build, test, deploy automatically
 - Infrastructure as Code: servers described in files
 - Why it is complex: security, costs, scaling, monitoring, many tools interacting
 - Why the profession is well paid: high responsibility and a shortage of experts

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Which topics interest you most? Let me know your first, second and third choice by October 23rd.

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